collegesofdistinction.com

How to Survive College Math

Nathan Wilgeroth

7–9 minutes

Of all subjects in school, math tends to be one of the most polarizing. We either love it, or we love to hate it. That passion only gets stronger as you take more challenging courses like calculus at the end of high school.

Is math not your thing? That's okay! We all have our strengths and weaknesses. And that's what's so great about college: you get to delve more into the area of study that you are most passionate about, unlike the more broad education you get in high school. But while math might not be something that you have to take through for *all* four years of college, it's still likely that you have to take at least *one* college-level math class. Most colleges and universities have a general curriculum, or common core, to ensure that you have a strong foundation of knowledge in multiple areas of study, including math.

We at Colleges of Distinction are no strangers to college math—and we're certainly no strangers to the fear that comes with *taking* college math. Get ready to survive college math with the help of our tips as well as the keen advice of real university professionals.

1. Read Your Syllabus

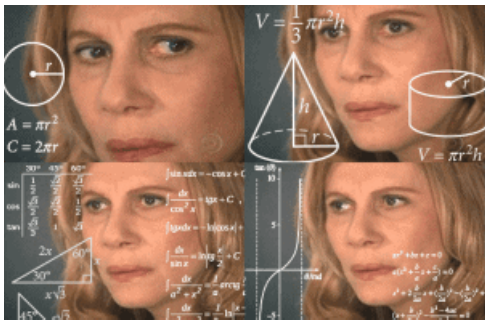
As with every college course, it's crucial for you to review your course syllabus both at the beginning and periodically throughout the semester. **Dr. Kim Ward** from Eastern Connecticut State University stresses the importance of understanding the rules and expectations set by your professor. "You should have clarity regarding how you will be evaluated in the course," she says. "There may be an attendance and class participation policy and required tutoring that affects your final course grade. You should know if there will be graded homework assignments, quizzes, tests, midterm and final exams. If so, you should know the dates of these assessments and how they are factored into your final course grade."

Featured School

2. Read, Period.

One of the starkest differences you might notice between high school and college math is that you will have a lot more independent work and learning to do. Instead of taking one class that meets about every other day for an entire year, college math courses typically meet two to three times a week and cover all the material within a single semester.

...



What?

Yeah, that's fast, which means your professor will likely ask you to read and comprehend much of your textbook on your own. Eastern's **Dr. Ward** outlines the many challenges that come with rigorous math courses. "The sequential nature of math coupled with its own vocabulary, need for persistent studying, and the speed at which math is taught in higher education, with approximately 15 weeks in a semester, creates major problems for college students." All of this mathematical jargon can be tough to retain, so it's important to keep up with your reading and highlight the important points and formulas you need to remember.

3. Think of Homework as Practice

It can be tempting to get through your homework as quickly as possible, working quickly to answer questions so that you can put it aside and enjoy all the antics you're looking forward to in college. If you're not particularly strong in math, however, you would be doing yourself a favor by putting lots of thought into the work you do outside of class. Practice enough problems so that you feel like you are in command of the material in front of you. And if you still feel a little shaky, you can look up more practice questions online. Sure, no one *loves* homework, but it is your opportunity to build your strength and understanding.

Dr. Jessica OShaughnessy, Dean of Student Success and Associate Mathematics Professor at Shenandoah University, has learned a lot about the way that students learn and retain information from both in the classroom and at home. "My daughter loves to read and sometimes asks me why she does better in English than math. I tell her that she practices reading every day for hours. If she spent nearly as much time on math, it would be her best subject too!" She advises her daughter and her students to practice homework as well as similar questions from the book. "Questions in the review sections of chapters are most useful," she adds, "as they often combine techniques."

4. Compound Skills and Notice Patterns

Don't get overwhelmed by all the concepts to which you'll be introduced in class. As **Dr. OShaughnessy** mentioned, review sections in your textbook often combine techniques and material from other units in the course. Use all of the information you learn to support you throughout the semester; there are usually patterns or procedures that show up again and again. Knowing this, you will be able to make connections that will help you remember older material and understand more quickly the new

material.

5. Seek Help During Office Hours

“Getting help during office hours, or making use of the university’s tutoring services, is not a sign of failure,” says **Dr. OShaughnessy**. “My best students are the ones that come to office hours and learn from their mistakes. Start early and come often.” Whether you have trouble with a specific problem, or if you just want help wrapping your head around a general concept, you will greatly benefit from every visit you take to your professor’s office hours.

Meeting one-on-one with your professor shows that you care about your work, and they will gladly help you in whatever way they can. Especially if you are enrolled in a large class, they won’t know that you’re struggling unless you ask for help. **Dr. OShaughnessy** says that “the biggest mistake students make is they realize they need help too late. If you get stuck on a homework problem, bring it to office hours, even if it is the first week of classes!”

In addition to asking for help outside of class, you will gain a lot of asking questions and being an active participant during class. Make a point to engage with your professor—raise your hand, ask for clarification, and let yourself engage with the material instead of sitting passively and letting the lecture wash over you. This will also motivate you to stay invested in the material by taking notes as you follow along.

[Read: Why You Shouldn’t Ignore the Benefits of Tutoring]

Featured School

6. In Practice: Chunk Tests/Quizzes

When it’s time for an exam, there are different strategies you can take to work efficiently under a time constraint. After you have studied and visited your professor’s office hours, you hopefully will have grasped which types of problems you rock at and which ones aren’t so solid. Skim through your entire exam before starting, as it will give you the opportunity to gauge how much time will be needed for each question. You will also be able to take note of any questions that might be weighed more heavily than others.

You don’t need to go through the questions in order, so plan your attack wisely! Give yourself the time to work thoughtfully through more challenging sections so that you aren’t rushed at the end of the exam period.

College courses are challenging no matter the subject, so these tools are applicable to any subject you

might be nervous about. Ultimately, your path to survival is paved by discipline and initiative. Stay on top of your readings, practice regularly and, perhaps most importantly, ask for help! When you become an active student, you will be able to grow and succeed. As Eastern's **Dr. Ward** puts it, "The culmination of enrolling in the appropriate first college math course, knowing the course expectations and policies, and addressing your affective characteristics is transitioning you to becoming a self-directed learner... Being an independent, self-directed learner is important to your academic success."